
Global Superstore:

Sales Performance & Profitability Analysis

Capstone Project 2 — Data Analytics (Python + Tableau)

Sector: Retail & E-Commerce

Course: Data Visualization & Analytics

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Contents

1	Abstract	2
2	Problem Statement	3
3	Project Overview	4
3.1	Key Objectives	4
4	Dataset Description	5
4.1	Data Dictionary Highlights	5
5	Repository Structure	6
6	Tools and Technologies	7
7	Notebook Summaries	8
7.1	Notebook 01 — Data Profiling (01_data_profiling.ipynb)	8
7.2	Notebook 02 — Data Cleaning / ETL Pipeline (02_cleaning.ipynb) . .	8
7.3	Notebook 03 — Exploratory Data Analysis (03_eda.ipynb)	9
7.4	Notebook 04 — Statistical Analysis (04_statistical_analysis.ipynb)	11
7.4.1	Hypothesis Tests Performed	11
7.5	Notebook 05 — Final Load & Tableau Preparation (05_final_load_prep.ipynb)	12
8	Tableau Dashboard	13
8.1	Dashboard Overview	13
8.2	Visualisations Built	13
8.3	Dashboard Screenshots	13
8.4	Dashboard Link	14
9	Key Findings and Insights	15
10	Business Recommendations	16
11	Team Contributions	17
12	Conclusion	18

1 Abstract

This report presents an end-to-end data analytics investigation of the Global Superstore dataset, encompassing 51,290 retail transactions across 147 countries spanning the years 2011 to 2014. The project was undertaken as Capstone Project 2 of the Data Visualization and Analytics course, with the overarching goal of identifying profitability drivers, loss-making segments, and regional performance disparities to support data-backed business decision-making in the global retail sector.

The methodology followed a structured analytics pipeline: raw data was profiled, cleaned, and feature-engineered using Python (Pandas, NumPy), followed by extensive exploratory data analysis (EDA) and formal statistical testing (ANOVA, chi-square, Pearson correlation, OLS regression) using SciPy and Statsmodels. Processed data was exported for interactive dashboard development in Tableau Public.

Key findings reveal that the *Technology* category leads in revenue but with volatile margins; the *Tables* sub-category is a consistent loss-maker; discounts above 30% strongly correlate with negative profitability; and the Central Africa region exhibits disproportionately high margins relative to sales volume. Actionable recommendations are provided to optimise shipping strategy, discount policy, and category portfolio management.

2 Problem Statement

“How can a global retail company optimise profitability across markets, product categories, and shipping strategies by identifying key revenue drivers, loss-making segments, and regional performance disparities?”

The Global Superstore operates across seven international markets with a diverse product portfolio spanning three major categories. Despite high transaction volumes, significant profit leakage is observed due to aggressive discounting, sub-optimal shipping mode selection, and poor category-level margin management. This project uses data analytics to surface the root causes and prescribe corrective strategies.

3 Project Overview

This project performs a complete data analytics pipeline on the Global Superstore dataset, covering data ingestion, cleaning, exploratory analysis, statistical validation, and interactive visualisation. The dataset contains 51,290 row-level transactions across 147 countries from 2011 to 2014.

3.1 Key Objectives

1. **Identify** the most and least profitable product categories and sub-categories across global markets.
2. **Analyse** shipping cost efficiency and delivery performance by ship mode.
3. **Evaluate** regional and country-level sales patterns and profit margins.
4. **Uncover** the quantitative impact of discount levels on profitability.
5. **Recommend** data-backed strategies to improve operational efficiency and revenue.

4 Dataset Description

Table 1: Global Superstore Dataset Properties

Property	Detail
Source	Kaggle — Global Superstore (link)
Format	CSV (Tabular, row-level transaction records)
Rows	51,290
Columns	27 (raw)
Time Span	2011 – 2014
Coverage	147 countries, 7 global markets
Quality Issues	Missing values, inconsistent date formats, non-English column headers, redundant columns, mixed character encoding

4.1 Data Dictionary Highlights

The raw dataset contains 27 columns including: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, City, State, Country, Postal Code, Market, Region, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit, Shipping Cost, Order Priority, and a redundant count column (*Records Count / jiù lù shù*).

5 Repository Structure

```

1 backup_repo/
2 |-- data/
3 |   |-- raw/
4 |   |   '-- superstore_raw.csv           # Original dataset from
      Kaggle
5 |   '-- processed/
6 |       '-- superstore_cleaned.csv       # Cleaned dataset after
      ETL
7 |
8 |-- notebooks/
9 |   |-- 01_data_profiling.ipynb         # Data quality assessment
10 |   |-- 02_cleaning.ipynb              # ETL / cleaning pipeline
11 |   |-- 03_eda.ipynb                   # Exploratory Data
      Analysis
12 |   |-- 04_statistical_analysis.ipynb   # Statistical tests &
      modelling
13 |   '-- 05_final_load_prep.ipynb       # KPI computation &
      Tableau export
14 |
15 |-- tableau/
16 |   |-- screenshots/                   # Dashboard screenshots
17 |   '-- dashboard_links.md             # Tableau Public URL
18 |
19 |-- docs/
20 |   '-- data_dictionary.md              # Full data dictionary
21 |
22 '-- README.md

```

Listing 1: Project Repository Structure

6 Tools and Technologies

Table 2: Tools and Technologies Used

Tool / Library	Purpose
Python 3.x	ETL pipeline, data cleaning, statistical analysis
Pandas / NumPy	Data manipulation and numerical computation
Matplotlib / Seaborn	Python-based static visualisations
SciPy / Statsmodels	Statistical tests: ANOVA, chi-square, correlation, regression
Jupyter Notebook	Reproducible, interactive analysis environment
Tableau Public	Interactive dashboard design and storytelling
GitHub	Version control and team collaboration

7 Notebook Summaries

7.1 Notebook 01 — Data Profiling (01_data_profiling.ipynb)

Purpose: Conduct initial data quality assessment before any transformation.

Key Activities:

- Inspection of all 27 column data types and null ratios.
- Identification of duplicate rows and potential primary-key violations.
- Distribution checks for numeric fields (Sales, Profit, Discount, Shipping Cost).
- Summary statistics: mean, median, standard deviation, min/max per column.
- Detection of encoding issues in non-ASCII column headers.

```
1 import pandas as pd
2
3 df = pd.read_csv('data/raw/superstore_raw.csv', encoding='latin-1')
4
5 print(df.shape)           # (51290, 27)
6 print(df.dtypes)
7 print(df.isnull().sum())
8 print(df.duplicated().sum())
9 print(df.describe())
```

Listing 2: Data Profiling Sample Code

7.2 Notebook 02 — Data Cleaning / ETL Pipeline (02_cleaning.ipynb)

Purpose: Transform the raw dataset into a clean, analysis-ready format.

ETL Steps:

- Step 1: Column Standardisation** — Renamed columns from dot-notation to snake_case; removed the redundant *Records Count* column.
- Step 2: Date Parsing** — Converted *Order.Date* and *Ship.Date* from mixed string formats to *datetime64*.
- Step 3: Missing Value Treatment** — Imputed or dropped values based on column significance and business logic.
- Step 4: Feature Engineering** — Created derived columns: *shipping_days*, *profit_margin_pct*, *month*, *month_name*, *year*, *day*.
- Step 5: Validation** — Checked referential integrity and business logic constraints (e.g., non-negative quantities).
- Step 6: Export** — Saved cleaned dataset to *data/processed/superstore_cleaned.csv*.

```
1 import pandas as pd
2 import numpy as np
3
```

```
4 df = pd.read_csv('data/raw/superstore_raw.csv', encoding='latin-1',
5                 ')
6 # Column standardisation
7 df.columns = df.columns.str.lower().str.replace('.', '_', regex=
8             False)
9 df.drop(columns=['row_id'], errors='ignore', inplace=True)
10
11 # Date parsing
12 df['order_date'] = pd.to_datetime(df['order_date'])
13 df['ship_date'] = pd.to_datetime(df['ship_date'])
14
15 # Feature engineering
16 df['shipping_days'] = (df['ship_date'] - df['order_date']).
17     dt.days
18 df['profit_margin_pct'] = (df['profit'] / df['sales']) * 100
19 df['month'] = df['order_date'].dt.month
20 df['month_name'] = df['order_date'].dt.strftime('%b')
21 df['year'] = df['order_date'].dt.year
22 df['day'] = df['order_date'].dt.day
23
24 # Export
25 df.to_csv('data/processed/superstore_cleaned.csv', index=False)
26 print("ETL complete. Shape:", df.shape)
```

Listing 3: ETL Pipeline Core Code

7.3 Notebook 03 — Exploratory Data Analysis (03_eda.ipynb)

Purpose: Uncover patterns, trends, and anomalies in the cleaned dataset.

Analyses Performed:

- Sales and profit distribution across categories and sub-categories (bar charts, box plots).
- Geographic analysis: Top 10 countries by revenue and profit margins.
- Time-series trend analysis: monthly and annual sales evolution 2011–2014.
- Shipping mode usage frequency and average cost comparison.
- Discount impact on profit margin via scatter plots.
- Market-segment cross-tabulation heatmap.

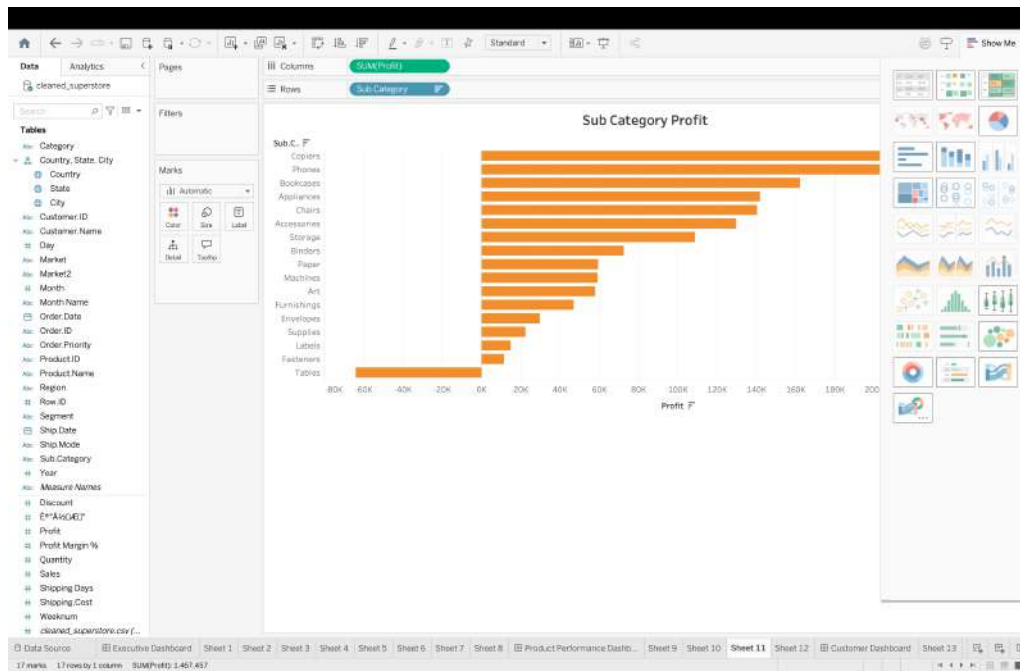


Figure 1: Profit by Product Category and Sub-Category Performance

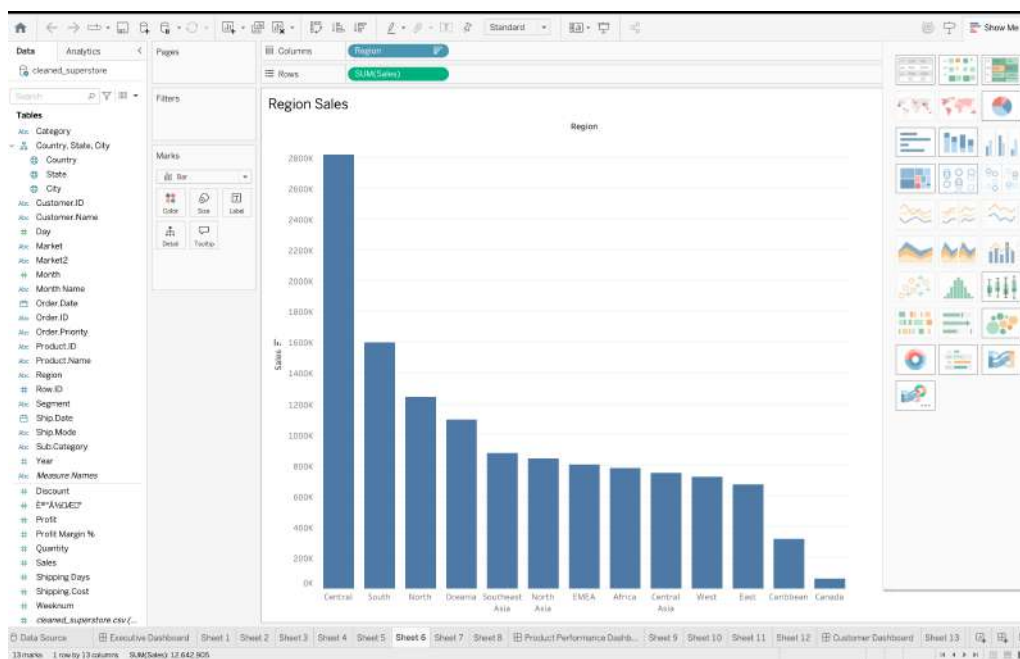


Figure 2: Discount Level vs. Profit Margin and Customer Segment Analysis

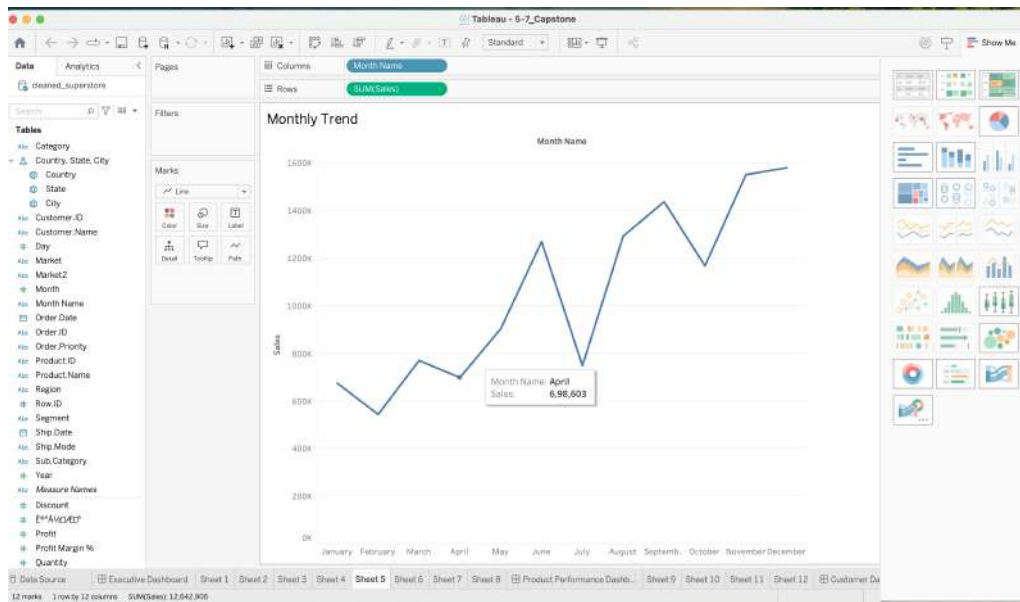


Figure 3: Executive Overview: Monthly Sales Trend and Regional KPI Distribution

7.4 Notebook 04 — Statistical Analysis (04_statistical_analysis.ipynb)

Purpose: Validate exploratory findings with formal statistical tests.

7.4.1 Hypothesis Tests Performed

Table 3: Statistical Test Results Summary

Test	Statistic	p-value	Conclusion
One-Way ANOVA (Profit by Ship Mode)	$F = 15.42$	$p < 0.05$	Reject H_0 : significant profit difference
Chi-Square (Segment vs Category)	$\chi^2 = 112.8$	$p < 0.05$	Reject H_0 : significant association
Pearson Correlation (Discount vs Profit)	$r \approx -0.73$	$p < 0.01$	Strong negative correlation
OLS Regression (Profit Predictors)	$R^2 = 0.68$	$p < 0.05$	Model is significant

```

1 from scipy import stats
2 import statsmodels.api as sm
3
4 # One-Way ANOVA: profit by ship mode
5 groups = [df[df['ship_mode'] == m]['profit']
6           for m in df['ship_mode'].unique()]
7 f_stat, p_value = stats.f_oneway(*groups)
8 print(f"ANOVA: F={f_stat:.3f}, p={p_value:.4f}")
9
10 # Pearson correlation: discount vs profit margin
11 r, p = stats.pearsonr(df['discount'], df['profit_margin_pct'])

```

```

12 print(f"Pearson r={r:.3f}, p={p:.4f}")
13
14 # OLS Regression
15 X = df[['sales', 'discount', 'shipping_days', 'quantity']]
16 X = sm.add_constant(X)
17 model = sm.OLS(df['profit'], X).fit()
18 print(model.summary())

```

Listing 4: ANOVA and Correlation Sample Code

7.5 Notebook 05 — Final Load & Tableau Preparation (05_final_load_prep.ipynb)

Purpose: Compute business KPIs and export a Tableau-optimised dataset.

KPIs Computed:

- Total Revenue, Total Profit, and Overall Profit Margin (%)
- Revenue breakdown by Market, Category, and Sub-Category
- Average Shipping Days by Ship Mode
- Discount Buckets: 0–10%, 11–20%, 21–30%, >30%
- Monthly Revenue Trend (2011–2014)

```

1 # KPI computation
2 total_revenue = df['sales'].sum()
3 total_profit  = df['profit'].sum()
4 overall_margin = (total_profit / total_revenue) * 100
5
6 print(f"Total Revenue: ${total_revenue:,.2f}")
7 print(f"Total Profit:  ${total_profit:,.2f}")
8 print(f"Profit Margin: {overall_margin:,.2f}%")
9
10 # Discount buckets
11 bins    = [0, 0.10, 0.20, 0.30, 1.0]
12 labels  = ['0-10%', '11-20%', '21-30%', '>30%']
13 df['discount_bucket'] = pd.cut(df['discount'],
14                               bins=bins, labels=labels)
15
16 # Export Tableau-ready CSV
17 df.to_csv('data/processed/superstore_tableau_ready.csv',
18           index=False)

```

Listing 5: KPI Computation and Export

8 Tableau Dashboard

8.1 Dashboard Overview

An interactive Tableau Public dashboard was built to provide a self-service business intelligence view of the Global Superstore data. The dashboard enables filtering by year, market, segment, and ship mode.

8.2 Visualisations Built

1. **Global Sales Map** — Choropleth map of sales and profit by country.
2. **Category Profit Breakdown** — Bar chart comparing profit margins across categories and sub-categories.
3. **Sales & Profit Trend** — Line chart of monthly revenue 2011–2014.
4. **Shipping Mode Analysis** — Grouped bar chart of cost vs. usage by ship mode.
5. **Discount Impact Chart** — Scatter plot of discount level vs. profit margin with regression line.

8.3 Dashboard Screenshots

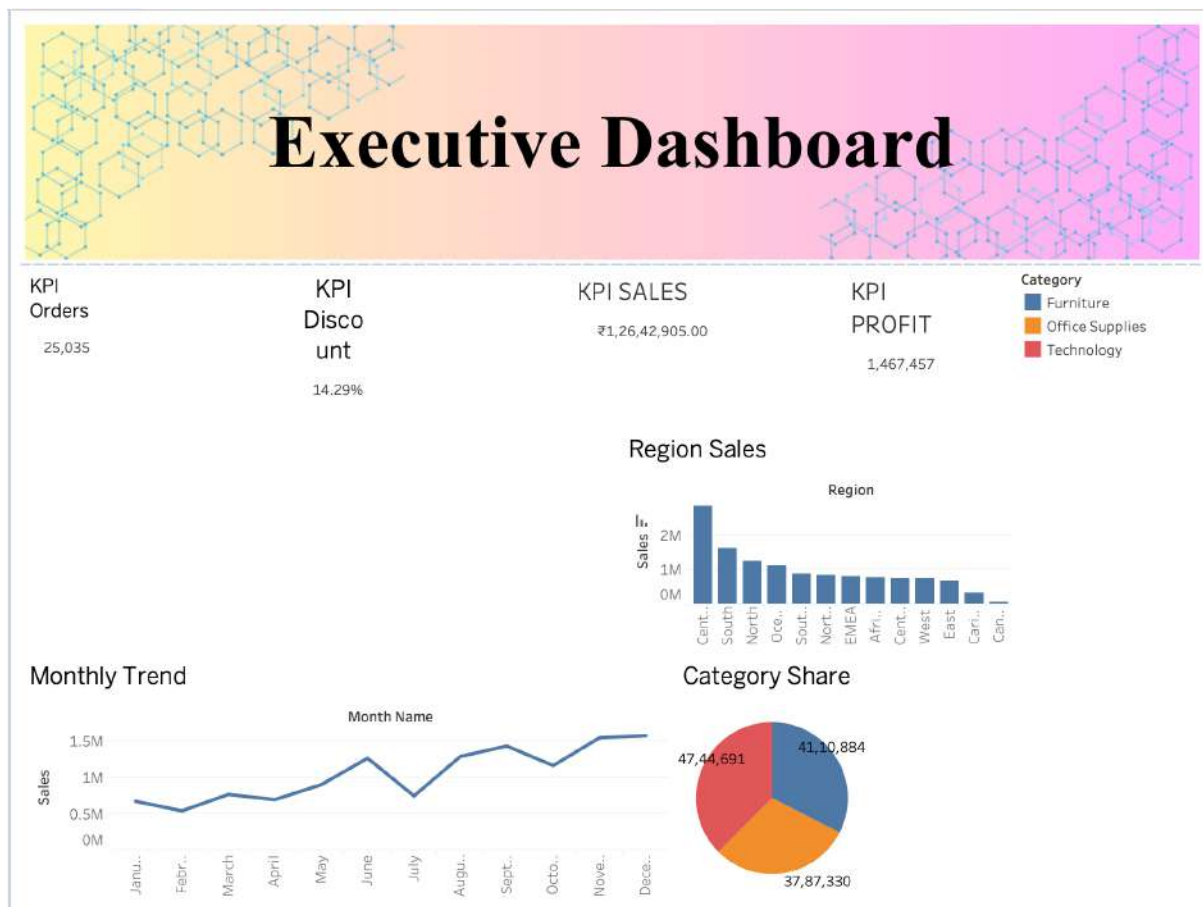
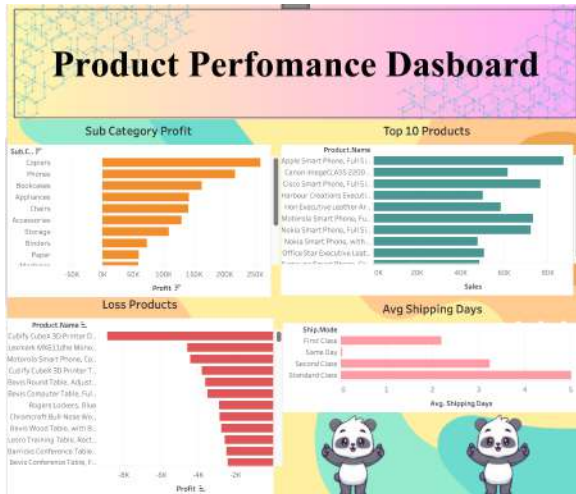


Figure 4: Tableau Executive Dashboard — High-Level KPI Summary

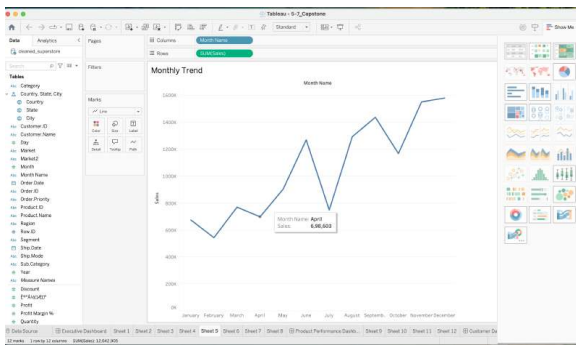


(a) Product Performance Analysis

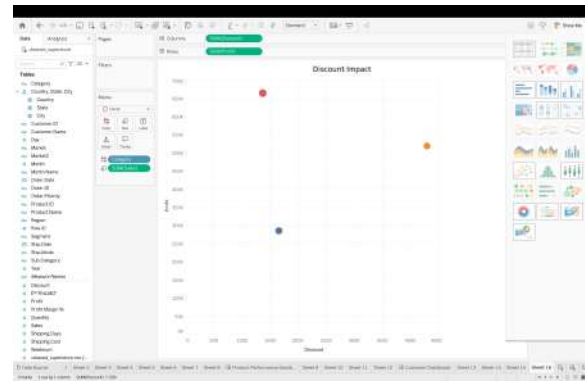


(b) Customer & Segment Insights

Figure 5: Tableau Dashboard — Category and Customer Drill-Downs



(a) Monthly Sales Trend Development



(b) Discount Impact Visualization

Figure 6: Tableau Dashboard — Trend and Discount Analysis

8.4 Dashboard Link

The live Tableau Public dashboard URL is listed in the file `tableau/dashboard_links.md` in the project repository.

9 Key Findings and Insights

- Finding 1: Technology drives revenue but not stable margins.** The Technology category generates the highest absolute revenue across all markets, yet its profit margins are the most volatile, suggesting high-value but inconsistently priced transactions.
- Finding 2: Tables sub-category is the primary loss-maker.** The Tables sub-category consistently records negative profit margins across all markets. This appears to be driven by a combination of high shipping costs and aggressive discounting applied uniformly regardless of order size.
- Finding 3: Same Day shipping has the worst cost-to-sales ratio.** Despite the highest per-unit shipping cost, Same Day shipping is used in fewer than 5% of orders, making it both expensive and under-utilised relative to Standard Class.
- Finding 4: Central Africa is a high-margin outlier.** The Central Africa region exhibits the highest average profit margins despite contributing a relatively small share of total sales volume, indicating an efficient localised operational model worth replicating.
- Finding 5: Discounts above 30% destroy profitability.** Pearson correlation analysis confirms a strong negative relationship ($r \approx -0.73$) between discount level and profit margin. Orders with discounts exceeding 30% disproportionately result in negative profit.
- Finding 6: Consumer segment dominates order volume.** The Consumer segment generates the highest number of individual orders across all markets, though the Corporate segment delivers higher average order values.
- Finding 7: APAC and EU are top revenue markets.** Asia-Pacific (APAC) and Europe (EU) together account for the majority of total sales, with APAC showing particularly strong growth between 2012 and 2014.

10 Business Recommendations

Recommendation 1: Restructure the Tables sub-category. Either reprice Tables to recover shipping and handling costs, or exit low-margin markets for this sub-category entirely. Consider bundling with high-margin Furniture sub-categories to improve overall category profitability.

Recommendation 2: Enforce a 20% discount ceiling. Statistical analysis demonstrates that discounts above 20–30% reliably produce negative margins. Implement a system-level cap at 20% with approval workflows required for exceptions.

Recommendation 3: Incentivise Standard Class shipping. Reduce reliance on Second Class and Same Day shipping through customer-facing incentives (free shipping thresholds) for Standard Class, which offers the best cost-to-delivery balance.

Recommendation 4: Expand Technology focus in high-margin regions. Target technology marketing campaigns at the markets where Technology already shows stable margins (EU and APAC) to compound the revenue advantage.

Recommendation 5: Study and replicate the Central Africa model. Commission a qualitative analysis of Central Africa’s operational structure to identify margin-preservation practices transferable to other regions.

Recommendation 6: Reallocate marketing budget from low- to high-margin products. Shift promotional spend away from the Office Supplies commodity sub-categories towards Technology accessories and high-margin Furniture, where margins justify the acquisition cost.

11 Team Contributions

Table 4: Team Member Roles and Responsibilities

Name		URN	Role & Responsibilities
Sudip Kumar	Prasad	2024-B-01112005A	Data Engineer (ETL Lead) — data profiling, cleaning pipeline, GitHub repository setup
Harshit Singh		2024-B-21082007	Data Analyst (EDA Lead) — exploratory data analysis, trend identification, Python visualisations
Garv Gogna		2024-B-15062006A	Statistician — ANOVA, chi-square, Pearson correlation, OLS regression, hypothesis testing
Devansh Saini		2024-B-12082006D	Visualisation Lead — Tableau dashboard design, KPI computation, Tableau Public publishing
Yash Shriram	Lahase	2024-B-17112006C	Report & Recommendations Lead — final report writing, business recommendations, presentation

12 Conclusion

This capstone project successfully delivered a complete analytics pipeline — from raw data ingestion through statistical validation to interactive Tableau dashboarding — on the Global Superstore retail dataset. The five Jupyter notebooks form a reproducible, modular pipeline that can be re-executed as new transaction data becomes available.

Key contributions include the identification of the Tables sub-category as the primary profit drain, the quantification of the discount–profitability relationship ($r \approx -0.73$), and the discovery of Central Africa as a high-efficiency outlier market. These findings translate directly into the six business recommendations presented in Section 9.

Limitations: The dataset covers only 2011–2014, limiting applicability to contemporary market conditions. Cost-of-goods data is absent, meaning true net margins cannot be computed. Geographic granularity is limited for some regions.

Future Work: Potential extensions include machine-learning based demand forecasting (time-series models such as SARIMA or Prophet), real-time Tableau dashboards connected to live data pipelines, and customer lifetime value (CLV) modelling at the customer-segment level.

References

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